



SKIN & MOOD UPLIFTER

GOLDEN AGE KEEPER

HEBELYS®



SKIN ISSUE STUDIED BY GREENTECH



SKIN & EMOTIONAL NEEDS OF SILVER GENERATION

According to demographic studies, 2020 will be a turning point: silver generation will outnumber children under age 5 (He *et al.*, 2016). The rise of the silver generation reflects human effort to increase longevity and health. In the same time, mature women's expectations have changed: they assume their age and its visible signs as well as they are more and more aware of the importance of an emotional wellbeing. From these trends a new dermo-cosmetic concept emerged: positive aging, which focuses on wellness to maintain health.

Besides well-known deleterious effect of aging on physiological processes (formation of senescent « zombie cells » and spread of senescence, matrisome alteration) (Scudellari, 2017; Freitas-Rodriguez *et al.*, 2017), aging has a negative effect on mood and self-esteem (Pearlman, 1993). Many women attach self esteem to their body image which is associated with beauty, femininity and youth.

As a pioneer in the field of positive aging, GREENTECH developed HEBELYS®, an advanced biotechnological active ingredient, designed using a psychobiological approach. HEBELYS® specifically addresses skin and emotional needs of the silver generation women and supports women after the age of 50, acting like a guardian of their golden age.

ACTIONS

MECHANISMS

PREVENTS & REDUCES SENESCENT « ZOMBIE CELLS »
REJUVENATES DERMAL MATRISOME & DERMIS INTEGRITY

SUBLIMATES

WRINKLES
TEXTURE
SUPPLENESS

AWAKES

SKIN RADIANCE
FRESHNESS &
EVENNESS

UPLIFTS

SELF-ESTEEM
& MOOD
(PSYCHOBIOLOGY)

MOLECULES

- GLYCOSPHINGOLIPIDS

SOURCING

SPHINGOMONAS HYDROPHOBICUM
GOLDEN EXTREMOPHILIC MICROORGANISM

Isolated from water samples
from the Silver Coast (France)
Original biotechnological process in
collaboration with Deinove.

« BIO-INSPIRATION FROM
EXTREMOPHILIC MICROORGANISMS »

COSMETIC USES

- POSITIVE AGING SKIN CARE
- PREMIUM DAILY SKIN CARE
- MATURE / SILVER GENERATION SKIN CARE
- PREVENTIVE CARE FOR WOMEN AGED 50+
- HOLISTIC ANTI-WRINKLE SKIN CARE



GREENTECH INNOVATION

> **ADVANCED BIOTECHNOLOGY**

PSYCHOBIOLOGICAL APPROACH
SCIENTIFIC PUBLICATION

> **TACKLES THE ROOT OF SKIN AGING**

PREVENTS & REDUCES SKIN ZOMBIE CELLS
LIMITS SENESCENCE SPREAD
REJUVENATES DERMAL MATRISOME

> **SKIN & MOOD UPLIFTER**

WOMEN'S GOLDEN AGE KEEPER

AGING IMPACTS

INITIATION:

- Aging/Exposome-induced signalling pathways: **p21** or **p16**

SENESCENT STATE = ZOMBIE CELLS:

- Damaged aged cells that stay alive... But don't work anymore!
- Aged skin *in vivo* : **20-60%** of «zombie fibroblasts».

SPREAD TO OTHER CELLS

- «Zombie cells» produce **SASP*** including inflammatory factors poisoning nearby cells and thus accelerating skin aging.

DERMAL MATRISOME

- Proteins related to the dermal extracellular matrix (ECM).
 - **Structural proteins**
 - **Adhesion proteins**
 - **Proteoglycans**
- A «young» matrix has a **rejuvenating effect on zombie cells**.

AGING
Glycation, aggregation, cross-linking

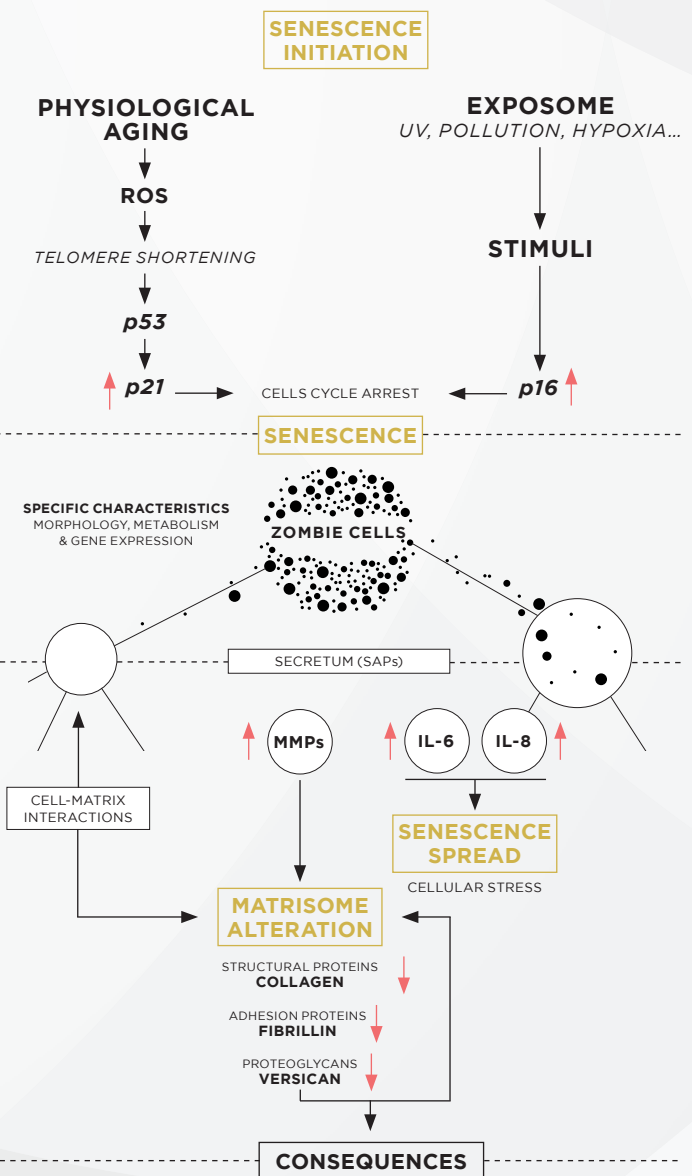
**TARGETING ZOMBIE CELLS & MATRISOME ALTERATION:
A WAY TO DELAY AGING**

PHYSICAL IMPACTS OF AGING

WRINKLES
DULL & UNEVEN SKIN TONE
COARSE SKIN TEXTURE
SAGGY SKIN

COGNITIVE IMPACTS OF AGING

LOSS OF SELF-ESTEEM
NEGATIVE MOOD



*SASP: Senescence-associated secretory phenotype
Herbig et al., 2006 ; Choi et al., 2011 ; Lewis et al., 2011 ; Scudellari, 2017 ; Ewald, 2019

PROVEN EFFICACY

IN TUBO/VITRO ANALYSIS

HEBELYS® OFFERS A COMPLETE APPROACH AGAINST CELL SENESCENCE

IN TUBO

In tube assay of lipid peroxidation (MDA) with HEBELYS® (1.2%)
**p<0.01 vs control

IN VITRO

On aged 3D full-thickness skin model, with HEBELYS® (0.1%) or retinoic acid (100nM) during 42 days. Results normalized by the total dermal area. ***p<0.001 vs control

AGING

p21

-58%***

p21 is a protein that induces cell damages leading to cell cycle arrest. It is a key factor in senescence.

PREVENTS
SENESCENCE INITIATION

OXIDATIVE STRESS

-34.5%**

EXPOSOME

p16

-20%***

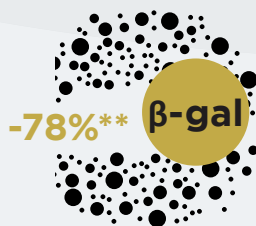
Better than retinoic acid
(p21 : -31%***; p16 : no effect)

p16 is a protein that limits cell turnover in dermis. It is a key factor in senescence and is thus involved in aging process

LIMITS THE ACTIVATION OF KEY SIGNALLING
PATHWAYS LEADING TO SENESCENT ZOMBIE CELLS

REDUCES ZOMBIE CELLS

β-galactosidase (β-gal) is an enzyme that enables to identify cells in senescence



Similar effect to retinoic acid (-76%**) (p21 : -31%***; p16 : no effect)

On aged 3D full-thickness skin model, with HEBELYS® (0.1%) or retinoic acid (100nM) during 42 days. Results normalized by the total dermal area. **p<0.01 vs control

LIMITS SENESCENCE SPREAD

Reduces the release of SASP-related inflammatory factors leading to cellular stress

IL-6

-33%

IL-8

-32%

On dermal fibroblasts (NHDF), H2O2 treatment, HEBELYS® (0.1%). Results normalized by the total protein concentrations. **p<0.01 vs control

HEBELYS® PROMOTES A HEALTHIER & REJUVENATING CELLULAR ENVIRONMENT

REJUVENATES DERMAL MATRISOME

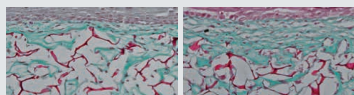
On aged 3D full-thickness skin model, with HEBELYS® (0.1%) or retinoic acid (100nM) during 42 days. Results normalized by the total dermal area. *p<0.05 vs control ; ***p<0.001 vs control

STRUCTURAL PROTEIN

Neocollagen I synthesis

+6%

Collagen I is the most abundant structural protein of dermal extracellular matrix and is related to skin resistance.



CONTROL

HEBELYS® 0.1%

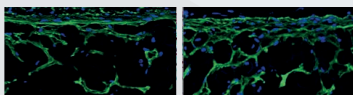
● Staining of collagen

ADHESION PROTEINS

Fibrillin expression

+22%***

Fibrillin allows the adhesion of ECM components and plays a key role in the production of cytokines and growth factors.



CONTROL

HEBELYS® 0.1%

● Staining of fibrillin

PROTEOGLYCANS

Versican expression

+7%*

Versican is linked to elastic fibers and promotes ECM organization. It has a key role in skin visco-elasticity.

CONTROL

HEBELYS 0.1%

+7%*

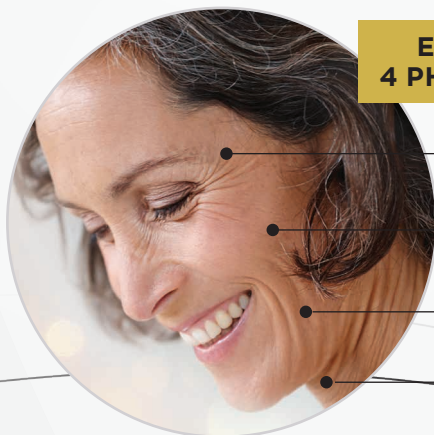
VERSICAN POSITIVE PERCENTAGE AREA IN DERMAL AREA

PROVEN EFFICACY

CLINICAL STUDY - IN VIVO

DESIGN

Double blind, open, intra-individual study. 24 silver generation women (60-70 yo).
Gel with **HEBELYS® 1%** vs placebo. Twice applications per day during **56 days**. Evaluations at D0 and D56.
Instrumental evaluations: Fringe projection: wrinkles, isotropy; Cutometer®; suppleness
Global visual effect on skin appearance standardized full face picture by VISIA®



EFFICACY EVALUATION ON 4 PHYSICAL IMPACTS OF AGING:

SMOOTHS

WRINKLES: DEPTH -8%*

(Crow's feet; roughness (Ra): -9%*; Depth (Rz): -8%*)

AWAKES

**DULL & UNEVEN SKIN TONE
LUMINOSITY 74% - UNIFORMITY 83%**

(self-assessment; % of agreement)

IMPROVES

COARSE SKIN TEXTURE +14%*

(Cheek; isotropy)

TIGHTENS

SAGGY SKIN +12%***

(Jaw; suppleness (Ue))

*p<0.05 vs control ; ***p<0.001 vs control

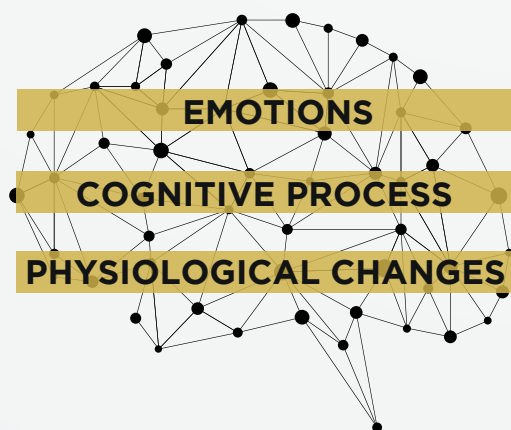


D0 - HEBELYS® 1%



D56 - HEBELYS® 1%

HEBELYS® OFFERS A VISIBLE SUBLIMATING
EFFECT ON MATURE SKIN



EFFICACY EVALUATION ON 2 COGNITIVE IMPACTS OF AGING

**2 SCIENTIFICALLY RECOGNIZED
PSYCHOLOGICAL TESTS**

SELF-ESTEEM

+10.5%***

(ROSENBERG TEST (ROSENBERG 1965))

MOOD STATE

+9%**

(PROFILE OF MOOD STATE (POMS) (MC NAIR 1971))

p<0.01; *p<0.001 D56 vs D0

« SKIN IS THE MIRROIR OF OUR PSYCHOLOGICAL STATE, THE MIRROR OF OUR EMOTION »

Anzieu (1974)

HEBELYS® HAS A MOOD UPLIFTING EFFECT

TECHNICAL INFORMATION**FORMULATION****Concentration for use:** 1%**pH for use:** 4.0 - 8.0**Caution for use:**Add to emulsions, at the end of the preparation process under 40 °C.
Formulation guidelines and stability studies available on demand**TECHNICAL DATA****Characteristics****Organoleptic** Appearance: liquid
 Colour: yellow to brown**Solubility** Water: soluble
 Glycols: partially soluble**Storage**

Keep in a dark place, in the original packaging, at an ambient temperature between 15° and 25 °C.

Tests of tolerance

- Skin irritation (tested at 2%): Non-irritant
- Sensitization (tested at 2%): very good skin compatibility
- Eye irritation (tested at 2%): partially non-irritant
- Phototoxicity: not phototoxic
- Mutagenicity test: Not mutagenic

• **INCI name:** *Propanediol, Sphingomonas ferment extract*• **Preservative:** None• **Authorized:** China**SCIENTIFIC ARTICLES**

- Gervason S, Napoli M, Dreux-Zigha A, Lazzarelli C, Garcier S, Briand A, Albory M, Thepot A, Berthon JY, Filaire E. Attenuation of negative effects of senescence in human skin using an extract from *S. hydrophobicum*: development of new skin care solution. *Cosmetics*, 2019.

- Gervason S, Dreux-Zhiga A, Demangeon J, Lazzarelli C, Berthon JY, Filaire E. A Psychobiological Approach for a Positive Skin Aging: Target Senescence Cells to Boost Emotion. *IFSCC Magazine*, 2020.

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